Weihang Guo

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Research Interests

World Models · Video Generation · Embodied AI · Robot Learning · Motion Planning · Robot Safety

Education

Rice University — Houston, TX
Ph.D. in Computer Science · Advised by Prof. Lydia E. Kavraki — *Aug. 2025 – May 2029 (expected)*

Rice University — Houston, TX
M.S. in Computer Science · GPA: 3.91/4.0 — *Aug. 2023 – Dec. 2024*

Lehigh University — Bethlehem, PA
B.S. in Mathematics and Computer Science · Dean's List all four years · GPA: 3.85/4.0 — *Aug. 2019 – May 2023*

Selected Publications (Google Scholar)

* Equal contribution † Mentoring

Agentic Robotics

[1] **CaStL: Constraints as Specifications through LLM Translation for Long-Horizon Task and Motion Planning**
W. Guo, Z. Kingston, and L.E. Kavraki
IEEE International Conference on Robotics and Automation (ICRA), 2025. [link]

[2] **Using VLM Reasoning to Constrain Task and Motion Planning**
M. Yan, M. Mengdibayev, A. Floros, **W. Guo**†, L.E. Kavraki, and Z. Kingston
Under review, 2025. [link]

[3] **Foundation-Model-Guided Topology-Aware Semantic Risk Fields for Manipulation**
G. Lee, **W. Guo**†, and L.E. Kavraki
Under review, 2026. [link]

Video Generation and Reasoning

[4] **Scaling Video Generation for Reasoning: At What Cost?**
W. Guo, X. Wu, Y. Wang, N. Mireshghallah, and L.E. Kavraki
Under review, 2026. [link]

[5] **Compress to Remember: Learning Compact Memory via On-Policy Distillation for Long Video Generation**
X. Wu*, **W. Guo***, Y. Wang*, X. Feng, L.E. Kavraki, and Z.S. Wu
Under review, 2026. [link]

Robot Learning and Safety

[6] **Zero-Shot Reactive Obstacle Avoidance for Generative Robot Policies**
W. Guo and L.E. Kavraki
Under review, 2026. [link]

Robotics Algorithms, Software, and Infrastructure

[7] **Efficient Multi-Robot Motion Planning for Manifold-Constrained Manipulators by Randomized Scheduling and Informed Path Generation**
W. Guo, Z. Kingston, K. Hang, and L.E. Kavraki
IEEE Robotics and Automation Letters (RA-L), 2026. [link]

[8] **The Open Motion Planning Library 2.0**
W. Guo, T. Tyrovouzis, E. Flores, C.W. Ramsey, Z.K. Kingston, I.A. Şucan, M. Moll, and L.E. Kavraki
IEEE Robotics & Automation Magazine (RA-M), to appear. [link] [GitHub] (2.2k stars)

[9] **Python Bindings for a Large C++ Robotics Library: The Case of OMPL**
W. Guo, T. Tyrovouzis, and L.E. Kavraki
IEEE International Conference on Robotics and Automation (ICRA), 2026. [link] [GitHub] (2.2k stars)

Selected Research Projects

Does Scaling Video Generation Lead to Reasoning? *2026 – Present*

- Conducted model, data, and compute scaling experiments [4] with seven Video DiTs (20M–938M parameters), trained from scratch on procedural Rubik's Cube videos to test history-dependent state reasoning.
- Showed that scaling improved state prediction but was computationally inefficient: symbolic guidance enabled a 20M model to match a 550M video-only model's performance with about 1/30 of the training compute.

Learning Compact Memory for Long-Horizon Video Diffusion *2026 – Present*

- Developed PACC [5], a memory-compression method that represents past frames with 10× fewer tokens, helping a pretrained video generator retain earlier visual information as videos grow longer without updating its weights.
- Improved MBench M-score by 15.4% relative (43.01 to 49.64) over the state-of-the-art baseline.

Training-Free Collision Avoidance for Generative Robot Manipulation Policies *2025 – 2026*

- Developed NUDGE [6], a model-agnostic, real-time safety method that injects signed-distance-field gradients into diffusion and flow-matching robot policies for whole-body collision avoidance without retraining.
- Improved 9 of 10 modified LIBERO tasks where 1-3 obstacles are added into the environment, and deployed a VLA ($\pi_{0.5}$) policy on a real UR5 with 2.46 ms latency per 50-step action chunk, which is 273× faster than the baseline.

LLM/VLM Agents for Long-Horizon Robot Task Planning *2024 – 2025*

- Developed CaStL [1], combining LLMs with formal-language planning for temporal reasoning over executable robot tasks with 10+ ordered steps and constraints, such as opening a container before placing an object inside.
- Co-developed a VLM-guided planner [2] that uses multimodal vision-language reasoning to identify spatial constraints, prune infeasible task plans before motion planning, and reduce failed planning branches.

Maintaining and Modernizing the Open Motion Planning Library (OMPL) *2024 – Present*

- Maintain OMPL [8], a cornerstone open-source library for sampling-based robot motion planning, and integrate modern planners, state spaces, and problem formulations for the robotics community.
- Replaced its legacy Python bindings with nanobind [9] for direct integration into Python-based robot-learning workflows and incorporated hardware-accelerated planners targeting millisecond-scale motion planning.

Technical Skills

Languages: Python, C/C++
Machine Learning and Generative Modeling: PyTorch, JAX, Hugging Face, DiTs, diffusion and flow matching, autoregressive video models, VAE-based latent tokenization, multimodal transformers, knowledge distillation
Robotics and Simulation: robot learning, real-world robot deployment, vision-language-action (VLA) model training, robot data collection, motion planning, OMPL, ROS, MuJoCo

Selected Honors and Awards

Jason Chahin Innovation Scholar, Rice University ($5,000)	*Feb 2025*
Donald J. Hillman Memorial Award in AI, Lehigh University	*Apr 2023*
2nd place on 10th F1TENTH Autonomous Grand Prix, ICRA 2022	*Jul 2022*
Mountaintop Research Fellowship, Lehigh University ($5,000)	*May 2022*

Teaching and Service

Teaching Assistant: Algorithmic Robotics, Computer Vision, IoT, and Autonomous Robotic Racing (Head TA)
Reviewing: ICRA 2025; IROS 2026; CoRL 2026; ICLR 2027; RA-L
Workshop Organization: Workshop on Open Robotics Software, IROS 2026
Invited Talk:
PhysCollaBot Workshop (IROS 2026) – *Efficient Motion Planning for Manifold-Constrained Multi-Robot Manipulators*. Presented on behalf of Prof. Lydia Kavraki.
Texas Regional Robotics Symposium (TEROS 2025) – *CaStL: Constraints as Specifications through LLM Translation for Long-Horizon Task and Motion Planning*.